Lecture 14 — Basis, dimension, and orthonormal vectors

Fri Sep 25

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Boyd video (optional): Lecture 16 — Gram–Schmidt · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Boyd 5.2–5.3 (basis, orthonormal vectors). Related objectives: Lectures. Schedule.

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples, and **Check yourself**. Gram–Schmidt is a short application at the end if time.

Learning objectives

By the end of class, you should be able to:

1. Explain what a basis accomplishes.
2. Identify a basis and dimension in simple examples.
3. Use linear combination, span, subspace, independence, basis, and dimension accurately.
4. Recognize an orthonormal collection.
5. Explain why orthonormal coordinates are convenient for measurement and reconstruction.

Warm-up

Let $a = (1, 0)$ and $b = (1, 1)$.

(a) Is $\{a, b\}$ independent? Does it span $\mathbb{R}^2$?

(b) Can every $v = (v_1, v_2)$ be written uniquely as $\alpha a + \beta b$? (Try $v = (2, 3)$.)

Fill in

A **basis** of a subspace V is a list of vectors that is both

1. _____ (no redundancy), and
2. a _____ set for V (every vector in V is a linear combination of the list).

What a basis accomplishes: every $v \in V$ has **coefficients** (coordinates) with respect to the basis — a unique recipe $v = \beta_1 a_1 + \cdots + \beta_k a_k$.

The **dimension** of V is the number of vectors in _____ of V. (All bases of V have the same length.)

Examples:

- $\dim \mathbb{R}^n =$ _____ (standard basis $e_1, \ldots, e_n$).
- A line through the origin through a nonzero a has dimension _____.
- A plane through the origin spanned by two independent vectors has dimension _____.

An **orthonormal** list $q_1, \dots, q_k$ satisfies $q_i^{\mathsf{T}} q_j =$ _____ if $i \neq j$, and $\|q_i\| =$ _____.

For an orthonormal basis of a subspace, the coefficients of v are especially simple:

$$\beta_i = _____$$

(inner products — no linear system to solve). That is why orthonormal coordinates are convenient for measurement and reconstruction.

Worked in class

W1. Basis or not? For each list, is it a basis of $\mathbb{R}^2$? If not, say whether it fails independence, spanning, or both.

(a) $(1,0),\ (0,1)$

(b) $(1,1),\ (2,2)$

(c) $(1,0),\ (0,1),\ (1,1)$

(d) $(1,1)$ alone

W2. Dimension. Let $V = \operatorname{span}\{(1,2,0),(2,4,0),(0,0,1)\}$. Find a basis of V and $\dim V$.

W3. Vocabulary drill. Fill each blank with one of: linear combination, span, subspace, independent, dependent, basis, dimension.

(a) $\operatorname{span}\{a, b\}$ is always a _____ of $\mathbb{R}^n$.

(b) If $c = 2a - b$, then $\{a, b, c\}$ is _____.

(c) A _____ of V is an independent spanning list for V.

W4. Orthonormal check. Which of these pairs in $\mathbb{R}^2$ is orthonormal?

(a) $(1, 0),\ (0, 1)$

(b) $(1, 1)/\sqrt{2},\ (1, -1)/\sqrt{2}$

(c) $(1, 0),\ (1, 1)$

W5. Orthonormal coordinates. Let $q_1 = (1, 0)$ and $q_2 = (0, 1)$ (standard basis — already orthonormal). For $v = (3, -2)$, what are $\beta_1 = q_1^{\mathsf{T}} v$ and $\beta_2 = q_2^{\mathsf{T}} v$? Check $v = \beta_1 q_1 + \beta_2 q_2$.

W6 (time permitting). Gram–Schmidt idea. From independent $a_1 = (1, 1)$ and $a_2 = (1, 0)$, form $q_1 = a_1/\|a_1\|$, then subtract from a_2 its projection onto q_1, and normalize. (Board sketch only.)

Check yourself

C1. A basis of a subspace is a list that is _____ and _____.

C2. $\dim \operatorname{span}\{(1,0,0),(0,1,0)\} =$ _____.

C3. True or false: every spanning list for V is a basis of V.

C4. If $\{q_1, q_2\}$ is orthonormal and $v = \beta_1 q_1 + \beta_2 q_2$, then $\beta_1 =$ _____.

C5. Why are orthonormal coordinates nicer than coordinates in a skewed basis like $(1, 0),\ (1, 1)$?

Answers (Check yourself)

C1. **independent** and **spanning** (for that subspace).

C2. 2

C3. **False.** Spanning lists can be dependent (too many vectors); a basis cannot.

C4. $\beta_1 = q_1^{\mathsf{T}} v$

C5. Coefficients are inner products (no 2×2 solve); lengths and angles stay easy because the basis vectors are unit and orthogonal.